Limits Graphically and Numerically

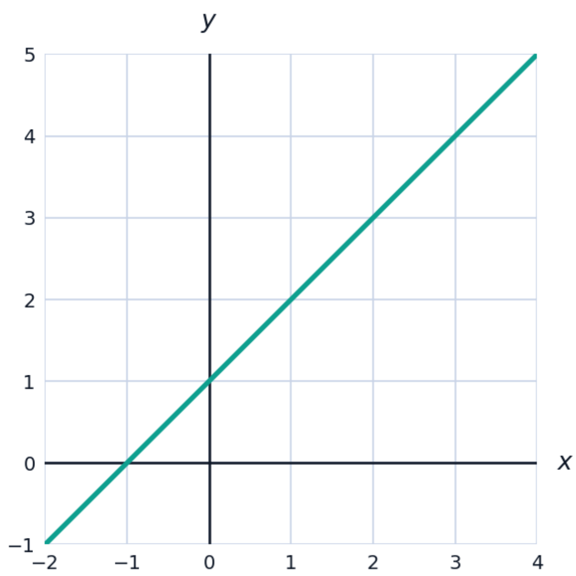
Estimating limits from a table

1 A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \to 3} g(x)$.

2 Using a table of values with x near 0 (radians), estimate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

Reading limits from a graph

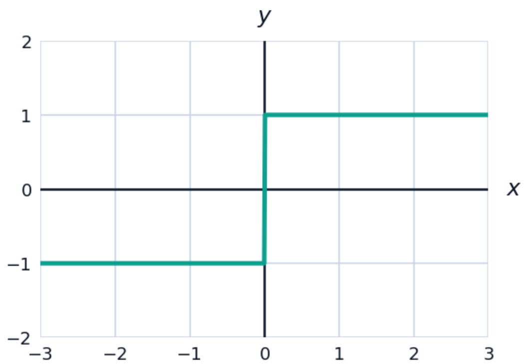
3 The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ $(x \neq 1)$, state $\lim_{x \to 1} f(x)$.



4 A function h has $\lim_{x \to 2^-} h(x) = 5$ and $\lim_{x \to 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \to 2} h(x)$ and explain whether h is continuous at $x = 2$.

One-sided limits and limits that fail to exist

5 For $f(x) = \frac{x}{|x|}$, find:



a $\lim_{x \to 0^+} f(x)$ **b** $\lim_{x \to 0^-} f(x)$ **c** $\lim_{x \to 0} f(x)$

6 A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

Algebraic evaluation

7 Evaluate each limit by direct substitution or factoring:

a $\lim_{x \to 4}(3x^2 - 5x + 1)$

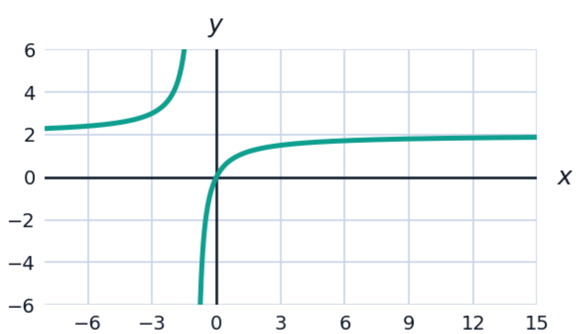
b $\lim_{x \to 3}\frac{x^2 - 9}{x - 3}$

c $\lim_{x \to -2}\frac{x^2 + 5x + 6}{x + 2}$

8 Evaluate $\lim_{x \to 0}\frac{\sqrt{x+9} - 3}{x}$ by rationalizing the numerator.

Limits at infinity from a graph

9 The graph shows $f(x) = \frac{2x}{x + 1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x \to \infty} f(x)$, then confirm algebraically by dividing numerator and denominator by x.



Limits involving infinite discontinuities

10 For $f(x) = \frac{1}{x-3}$, find $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$, and state whether $\lim_{x \to 3} f(x)$ exists.

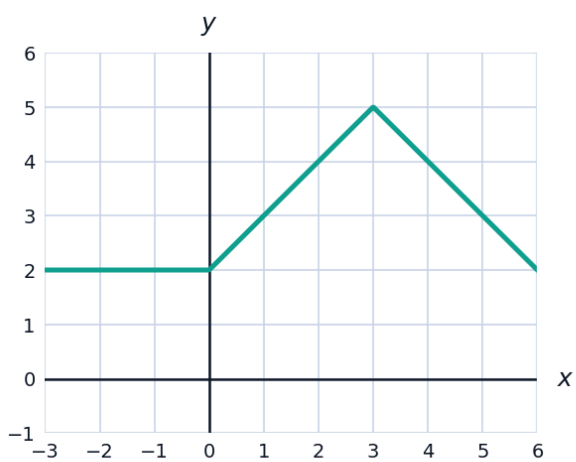
11 Evaluate $\lim_{x \to \infty} \frac{3x^2+1}{x^2-4}$, using the fact that the numerator and denominator have the same degree.

Estimating limits with a calculator-style table

12 Construct a mental table of values for $f(x) = \frac{e^x - 1}{x}$ at $x = -0.1, -0.01, 0.01, 0.1$ (all close to 1), and use it to estimate $\lim_{x \to 0} \frac{e^x - 1}{x}$.

Limits with piecewise graphical behavior

13 A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \le x < 3$, and $f(x) = 8 - x$ for $x \ge 3$. Find $\lim_{x \to 0} f(x)$ and $\lim_{x \to 3} f(x)$, stating whether each exists.



14 Evaluate $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.